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MILITARY SPECIFICATION

SHIELD, STOVEPIPE, TENT, NONMETALLIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a flexible, silicone rubber coated shield for a tent stovepipe opening.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-C-10797	-	Cloth, Coated, Glass, Silicone Rubber-Coated

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8340

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

8-1-10 - Shield, Stovepipe, Tent, Nonmetallic; Assembly and Details

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Tensile Strength
D 573 - Air Heat Test, Air Heating Medium
D 1053 - Stiffness Torsional Low-Temperature
D 2240 - Hardness Durometer
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide samples. Samples when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

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3.2 Materials and components. Materials (see 6.5) and components shall be as specified on the applicable drawings and as specified herein.

3.2.1 Coated cloth. The coated cloth for making the main piece of the shield shall be silicone rubber coated glass cloth conforming to MIL-C-10797.

3.2.2 Rubber. The rubber for the molded collar shall be compounded to conform to the requirements specified in table I and shall be pigmented such that the color approximates Olive Drab 210.

TABLE I. Physical requirements for rubber of molded collar

Characteristic	Requirements
Tensile strength, pounds per square inch (minimum):	
Before aging	600
After aging	400
Ultimate elongation, percent (minimum):	
Before aging	175
After aging	125
Hardness, scale reading:	
Before aging	60 ± 5
After aging, maximum increase, percent	25
Cold resistance at -85°F, minimum angular twist, degrees	1/

1/ The angular twist of the specimens in degrees shall be not less than that specified for the specimen thickness in ASTM D 1053.

3.3 Construction. At the option of the contractor, the construction of the shield shall be in accordance with construction A or B as specified in 3.3.1 and 3.3.2.

3.3.1 Construction A. The shield shall consist of two parts: the main piece and the molded collar. The dimensions of the shield shall be as specified in Drawing 8-1-10 and the cross-sectional configuration shall be as specified for construction A in Drawing 8-1-10. The main piece shall be made of the coated cloth specified in 3.2.1. The collar shall be made of the rubber specified in 3.2.2. The hole edge of the main piece shall be inserted into the rubber collar. The collar shall then be stitched to the main piece with one continuous row of stitching 1/8 to 3/16 inch from the outside edge of the

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collar. Stitching shall be stitch type 301, 4 to 5 stitches peer inch. The thread shall be as specified in 3.3.1.1. The last end of the stitching shall be run over the first end for a distance of 1 to 1-1/2 inches. Thread ends shall be trimmed to not more than 1/8 inch.

3.3.1.1 Thread. The thread for stitching the collar to the main piece shall be polytetrafluoroethylene (TFE) coated glass thread natural color, equal to or better than Dodge Industries, Fluoroglass Thread M753-18. The thread shall meet the following requirements (see 4.2.1):

Breaking strength, pounds	-	15 minimum
Length per pound, yards	-	1800 minimum, 2400 maximum
TFE content, percent by weight	-	12 minimum

3.3.2 Construction B. The shield shall consist of two parts; the main piece and the molded collar. The dimensions of the shield shall be as specified in Drawing 8-1-10 and the cross-sectional configuration shall be as specified for construction B in Drawing 8-1-10. The main piece shall be made of the coated cloth specified in 3.2.1. The collar shall be made of the rubber specified in 3.2.2. The collar shall be molded by the means of vulcanizing over the main piece and there shall be no visible evidence of cloth showing on the inside edge of the molded collar.

3.4 Strength of collar. There shall be no separation of the collar from the main piece or tearing away of the rubber in the collar when tested as specified in 4.2.5.

3.5 Identification label. Each shield shall be permanently and legibly marked along the edge on the back or reverse side with black ink in letters not less than 1/2 inch in height with the following information:

Stock Number
Nomenclature
Contract Number
Date of Manufacture

3.6 Workmanship. The finished shields shall conform to the quality and grade of product established by this document and the occurrence of defects shall not exceed the specified acceptable quality levels in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection require-

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ments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document. A certificate of compliance shall be furnished with each shipment or lot stating that the materials and components conform to the applicable requirements listed below:

Material component	Requirements
Thread (composition, breaking strength 1/ length per pound, TFE content)	3.3.1.1

1/ When tested in accordance with Method 4100 of FED-STD-191.

4.2.2 Rubber test. In addition to the requirements of the applicable referenced documents the rubber shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of one shield. The sample unit shall be four pieces of the rubber compound for the collar, molded and vulcanized under the same conditions as the finished shields. Each piece shall be 6 by 6 inches by 0.075 $\pm$ 0.010 inch thick. The sample size (number of sample units) shall be as specified below and the lot shall be unacceptable if any sample unit fails to meet any requirement specified. A certificate of compliance shall be furnished with each shipment or lot stating that the test pieces of the rubber compound were molded and vulcanized under the same conditions as the finished shield.

Lot size (shields)	Sample size (sample units)
800 or less	2
801 to 22,000	3
22,001 or more	5

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TABLE II. Rubber tests

Characteristic	Requirement paragraph	Test methods in American Society for Testing and Materials
Tensile strength before aging	Table I	D 412 1/
Tensile strength after aging	Table I	D 412 1/ and D 573 2/
Ultimate elongation before aging	Table I	D 412 1/
Ultimate elongation after aging	Table I	D 412 1/ and D 573 2/
Hardness before aging	Table I	D 2240 3/
Hardness after aging	Table I	D 2240 3/ and D 573 2/
Cold resistance	Table I	D 1053 4/

1/ Die No. III shall be used to cut test specimens.

2/ Except that test specimens shall be aged for 3 hours at 600°F.

3/ The Shore-A-Durometer shall be used to measure the hardness.

4/ The temperature of the exposure shall be minus 85°F and the heat transfer medium shall be methanol. The exposure time shall be 10 minutes.

4.2.3 End item visual examination. The end item shall be examined for the defects listed in table III, and applicable to the construction (A or B) furnished. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Main piece	Tear, cut, or hole (excepting pinhole).	X	
	Abrasion, resulting in thin or weak place or exposure of uncoated cloth structure.	X	
	Blister or soft lump.	X	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Main piece (cont'd)	Hard lump exceeding 1/2 inch in combined length and width directions, clearly noticeable.		X
	Crease or wrinkle resulting in doubling that cannot be corrected by manual pressure or adhesion of surfaces against each other.	X	
	Area devoid of coating, measuring more than 1 inch in any lineal direction.	X	
	Area devoid of coating, measuring 1 inch or less in any lineal direction.		X
	Color not OD-209.		X
	Coating tacky.	X	
Rubber collar	Crack, check, pit, or blister.	X	
	Tear, void, sunburst, lump, or shrinkage mark.	X	
	Color of molded rubber collar does not approximate OD-210.		X
	Shade does not match approved guide sample.		X
Thread	Not natural color.		X
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein).	X	
Construction and workmanship (general)	Any component missing.	X	
	Edge of cloth visible through inside edge of molded collar.	X	
	Wrong stitch type.	X	
	Stitches per inch less than 4 or more than 5.	X	
	Stitching margin less than 1/8 or more than 1/4 inch.	X	
	Stitching margin more than 3/16, but not more than 1/4 inch.		X
	Overstitching of first end: Less than 1 inch but not less than 1/2 inch or more than 1-1/2 inches.		X

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (cont'd)	Overstitching of first end: (cont'd)		
	Less than 1/2 inch.	X	
	Thread ends not trimmed to within 1/8 inch.		X
	Broken stitches and skip stitches:		
	More than one consecutively or more than three per shield.	X	
	More than one but not more than three provided the broken stitches are at least 3 inches apart.		X
	Run off.	X	
	Tight stitching showing obvious splitting of rubber collar.	X	
	Loose stitching, thread loops on top or bottom surfaces which can be entered by a size 18 needle.	X	
	Delamination or separation of collar from main piece:		
	- Greater than 1 inch.	X	
	- 1 inch or less.		X
	Appreciable amount of rubber from molded collar distributed on sides of reinforcing rings seriously affecting appearance.		X
	Malformation, irregularity, or poor trim of molded collar.	X	
	Spot, stain, streak, or other discoloration clearly visible.		X
	Overall uncleanness.		X
	Objectionable odor.		X
NOTE:	Aromatic odors usually associated with specific types of rubber compounds shall not be regarded as objectionable.		
Marking	Smeared, missing, incomplete, incorrect, or illegible.	X	
	Not in proper location.		X
	Height of lettering not as specified.		X
	Not permanent, i.e., can be easily rubbed off with moistened thumb.		X

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4.2.4 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.5 End item testing. The end item shall be tested as specified in 4.3.1 for conformance to the collar strength requirements specified in 3.4. The lot size shall be expressed in units of shields. The sample size shall be the number of shields indicated by inspection level S-2. Any test failure shall be cause for rejection of the lot.

4.2.6 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred, shall be 2.5.

Examine	Defect
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Contents	Number of unit packs per shipping container is more or less than required. Number of shields per unit pack is more or less than required. 1/

1/ For this defect, one bundle from each shipping container in the sample shall be examined.

4.2.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.3 Methods of inspection.

4.3.1 Strength of collar test. The strength of the collar shall be determined as follows:

Specimen - One shield. Two determinations shall be made on each shield.

Apparatus - Tensile testing machine specified in ASTM D 412 except as follows:

A hand vise clamp with jaws 1-1/2 ± 1/16 inches wide (Millers Falls No. 3 or equivalent) shall be attached to the upper grip. The lower grip shall be a clamp with jaws measuring 5-1/2 inches ± 1/8 inch by 1-1/4 ± 1/8 inches. The rate of travel for the power actuated grip shall be 2.0 inches + 0.1 inch per minute.

Construction A. Insert the main piece of this shield between the upper gripping surfaces so that the edge of the collar is just above the gripping surfaces and the edge of the shield closest to the gripped part is perpendicular to the direction of travel. Clamp that edge of this shield in the lower grip and adjust the upper grip opening to 3/64 ± 1/64 inch. (see illustration for testing on Drawing 8-1-10 typical for A or B construction). The second determination shall be made with the shield in a position 90° to that of the first.

Construction B. Insert the thin portion of the rubber collar between the upper gripping surfaces so that the thick portion of the collar is just above the gripping surfaces and the edge of the shield closest to the gripped part of the collar is perpendicular to the direction of travel. Clamp the edge of the shield in the lower grip and adjust the upper grip opening to 3/16 ± 1/64 inch (see illustration for testing on Drawing 8-1-10 typical for A or B construction). The second determination shall be made with the shield in a position 90° to that of the first.

Report - The specimen shall be considered as failing with respect to bond strength if failure occurs in either of the two determinations. Test failure is defined as separation of the collar from the main piece or tearing away of the rubber in the collar to the extent that application of 100 pounds

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of force cannot be attained. When a result of less than 100 pounds is caused by other than the two stated conditions, that determination will be considered invalid and additional determination shall be performed.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Ten shields shall be evenly stacked on a fiberboard pad that measures approximately 18-1/4 by 18-1/4 inches. The pad, in any case, shall extend one-eighth of an inch beyond the bottom shield at all edges. The stacked shields shall be securely crosstied with cotton twine to form a bundle. The bundle shall be packed flat in a fiberboard box conforming to style RSC, type CF, variety SW, class domestic, grade 200 of PPP-B-636. The fiberboard pad shall be made of the same material as the box. Inside dimensions of the box shall be 18-1/2 inches in length, 18-1/2 inches in width and 4-1/2 inches in depth. Box closure shall be effected with a 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. Shields shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Fifty shields, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, five in depth within a shipping container. The dimensions of each shipping container shall approximate 19-1/4 inches in length, 19-1/4 inches in width and 25 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.6.

5.2.2 Level B packing. Fifty shields, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, five in depth within a shipping container. The inside dimensions of each shipping container shall approximate 19-1/4 inches in length, 19-1/4 inches in width, and 25 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.6.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.6.

5.2.3 Commercial packing. Shields preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shields packed as specified in 5.2 shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129, or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The stovepipe shields are for installation in a tent roof through which a stovepipe passes. The shields act as insulation to protect the tent and canvas against excessive heat.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
c. When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.1).
d. When palletization is required (see 5.3).

6.3 Sample. For access to sample, address inquiry to the contracting activity issuing the invitation for bid (see 3.1).

6.4 Rubber composition. A silicone rubber composition has been found to satisfactorily meet the requirements of table I (see 3.2.2).

6.5 Recycled materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.7 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement number STANAG 2311. When amendment, revision, or cancellation of this document is proposed which will effect or violate the international agreement concerned, the preparing activity will take the appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activities:

Navy - MC
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